

Anything can be elided if you know how:
clausal ellipsis without identity

Till Poppels and Andy Kehler

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—I did (know that VPs can be elided as well).
 - (3) **Sluicing:** Joe was murdered but we don't know how (he was murdered).
- There are many other types of ellipsis.
 - This talk focuses on **sluicing**: ellipsis of clauses that are embedded under interrogative wh-phrases, as shown in (3).

Theories of ellipsis

A running example

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"No New Words" explanation

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Central claim: Material can only be elided if it is identical to its antecedent.

(e.g. Chung, 2006, 2013; Merchant, 2001; Rudin, 2018; Sag, 1976)

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Referential theories of ellipsis

Central claim: Ellipsis involves a null pro-form that presupposes that its referent is topical and uniquely identifiable in common ground.

(e.g. Hardt, 1993; Kehler, 2019; Poppels & Kehler, 2017; Webber, 1978)

Evidence that ellipsis involves discourse reference

Table: Diagnosing anaphoricity in ellipsis

	Pronouns	NPE	VPE	Sluicing	Gapping
Cataphora					
Exophora					
Non-local antecedents					
"Split" antecedents					
Inferred referents					
"Sloppy" readings					

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(Hardt, 1993; Kehler, 2019; Poppels & Kehler, 2017, 2019a, 2019b;
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Plan for today

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Summary of theoretical backdrop

- 2 theories of ellipsis: IDENTITY and Discourse Reference
- (7) appears to favor IDENTITY theories
- However: independent reasons to hold on to referential theory
- **Goal for today:** re-evaluate both ID-based explanations of (7)
 - Expts 1-3 test the **No New Words** explanation
 - Expt 4 tests the **Voice Mismatch** explanation

The “No New Words” Generalization

Chung (2006):

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- (8) a. Joe was murdered but we don't know who #(he was murdered **by**).
b. Joe was murdered but we don't know by who (he was murdered).
- (9) a. Joe is jealous but he didn't say who #(he was jealous **of**).
b. Joe is jealous of someone but he didn't say who (he was jealous of).

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- Ironically, Chung (2013) provides a counterexample in the title:

(10) Syntactic identity in sluicing: how much and why

Expt 1: Stimuli

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10 classic minimal pairs from the literature

- (11) a. Joe is jealous. —Do we know who? [Classic Bad]
b. Joe is jealous of someone. —Do we know who? [Classic Good]

⇒ **NNW prediction**: clear bimodal distribution in acceptability.

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(13) Can I borrow your textbook? —Why?

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20 unrelated fillers (distraction from manipulation)

- (14) a. The jug broke. —Any idea who? [Bad filler]
b. You didn't answer my question. —Which question? [Good filler]

Expt 1: Acceptability task

A: Do you mind if I put on some music?

B: **What kind of music?**

*(Consider what the **second speaker** says. Is this an acceptable English sentence/question in this context?)*

(unacceptable)

(fully acceptable)

Use number keys or click boxes to answer.

Expt 1: Paraphrase task

A: Do you mind if I put on some music?

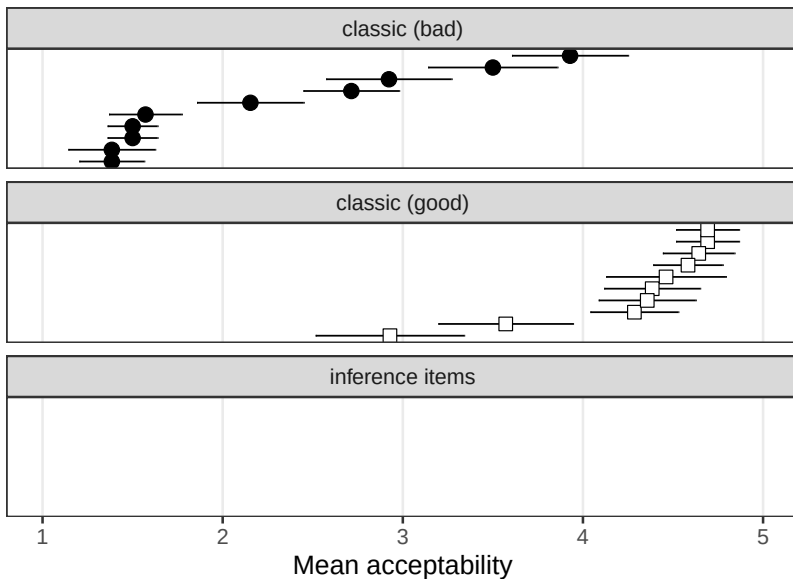
B: **What kind of music?**

*(What exactly does the **second speaker** mean?)*

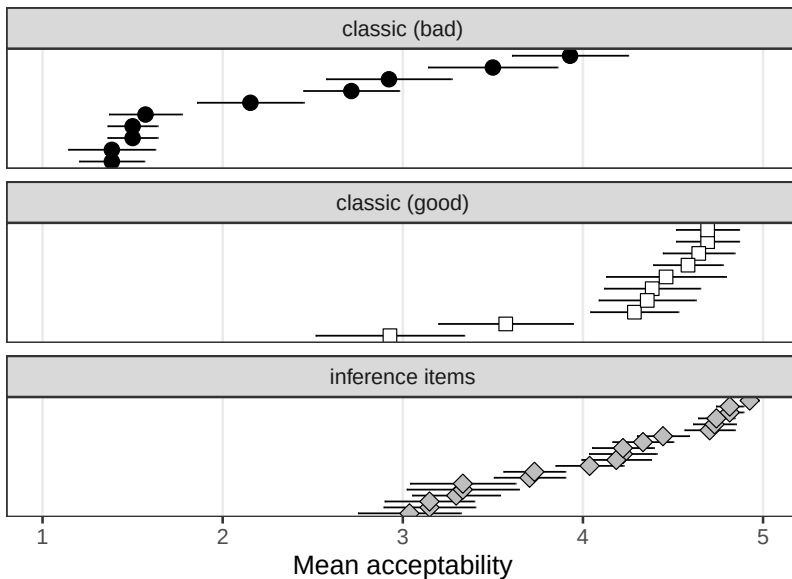
B: "What kind of music ?"

→ [Click here to continue](#)

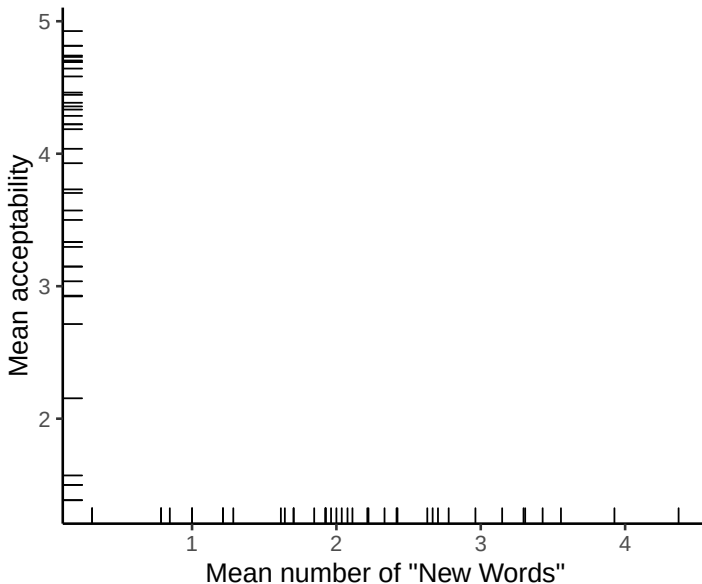
Results: acceptability



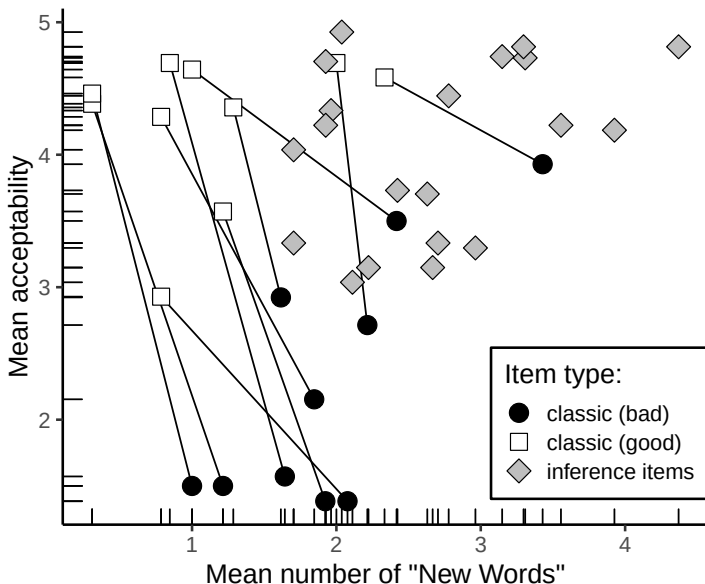
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Results (cont'd): # of "New Words"



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Expts 2 and 3: Sluices with nominal antecedents



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- (15) Did you not tell your friends about the game today? —I did, but I forgot to tell them where (the game would be; it would take place; ...).

Expts 2 and 3: Sluices with nominal antecedents



(15) Did you not tell your friends about the game today? —I did, but I forgot to tell them where (the game would be; it would take place; ...).

- Nominal-antecedent sluices are problematic for IDENTITY theories.
- Only previous mention: Beecher (2007).

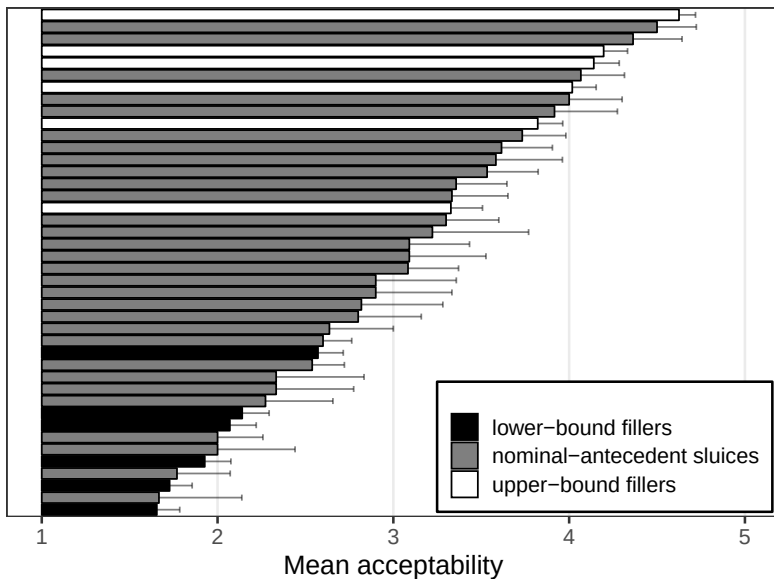
Expt 2: Stimuli

(16) A: I can't see your parents in the audience. Did you not tell them about your performance today?

B: I did, but I forgot to tell them {when|where|what about|how long|why}.

- 30 sluices with nominal antecedents like (16):
 - 6 contexts
 - 5 wh-phrases
- 12 fillers
- 63 participants

Results (Expt 2)



Expt 3

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Expt 3

- **Question:** can we explain this gradability?
- Challenging for IDENTITY theories
- QUD theories: sluicing depends on the availability of a relevant Question under Discussion (QUD) (AnderBois, 2014; Barros, 2014; Roberts, 2012)
- Plan for Expt 3:
 - Measure predictability of sluiced question (QUD)
 - Check if this ‘QUD availability’ measure predicts acceptability

Expt 3: passage-completion task

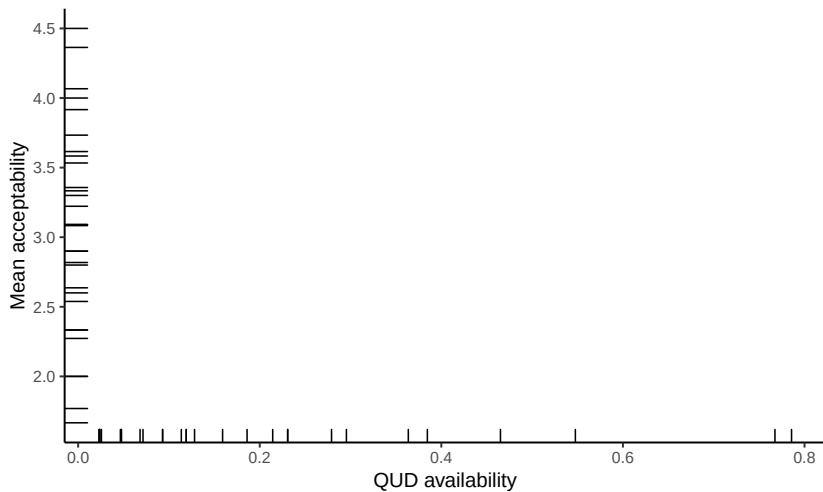
(Instructions: carefully read the passage below, carefully read all continuations, and then choose the continuation you find most likely.)

"A: I can't see your parents in the audience. Did you not tell them about your performance today? B: I did, but I forgot to tell them...

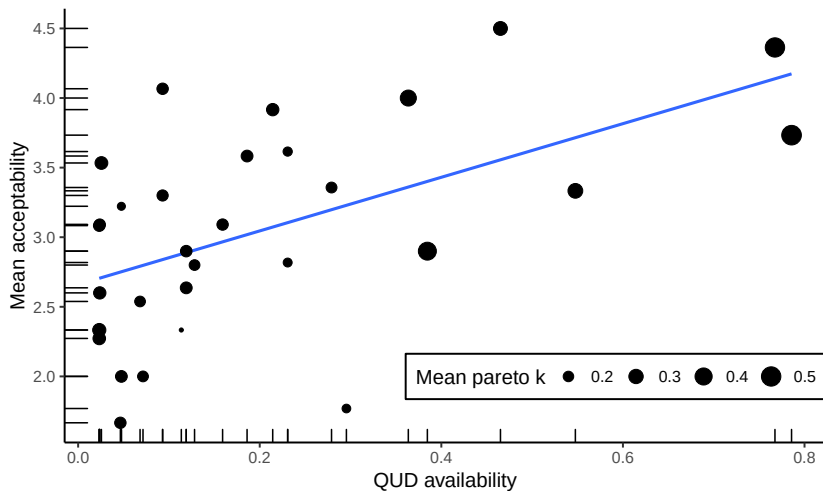
1. ...where I was performing."
2. ...how long it would last."
3. ...when it was going to start."
4. ...why they should be in the audience."
5. ...what it was about."

- Answer choices: established in separate norming experiment (paraphrase task analogous to Expt 1)

Results: QUD availability



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- Frequentist analysis: $\beta = 1.826, p = 0.037$
- Bayesian analysis: $P(\beta > 0) = 0.983$

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- Sluices with nominal antecedents are possible.
- Some are impeccable, some are terrible.
- Some of the variance can be explained by ‘QUD availability’.
- Caveat: most of the variance remains unexplained.
- Another caveat: haven’t yet tested unelided controls.

Sluicing and Voice Mismatches

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a. #(murdered him).

Voice mismatch explanation

b. #(he was murdered **by**).

“No New Words” explanation

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Voice mismatches matter beyond this example (anonymous CLS53 reviewer):

*[A]s Merchant 2013 (LI) showed, the major explicandum in the area of “mismatches” is the fact that **VPE**, but not **sluicing**, allows for voice mismatches.*

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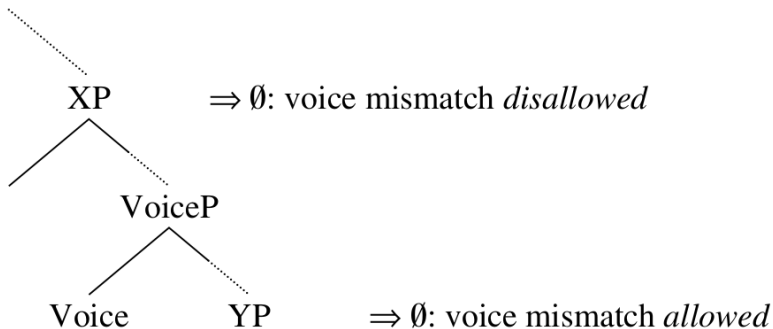
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From Merchant (2013, p. 78):

*The generalization that emerges is that when the target of ellipsis is a small amount of structure, such as a **VP**, mismatches in voice appear to be possible, but when more structure is targeted, as in **sluicing** and the like, no voice mismatch is allowed.*

Sluicing and Voice Mismatches

Merchant (2013, Fig. 1):



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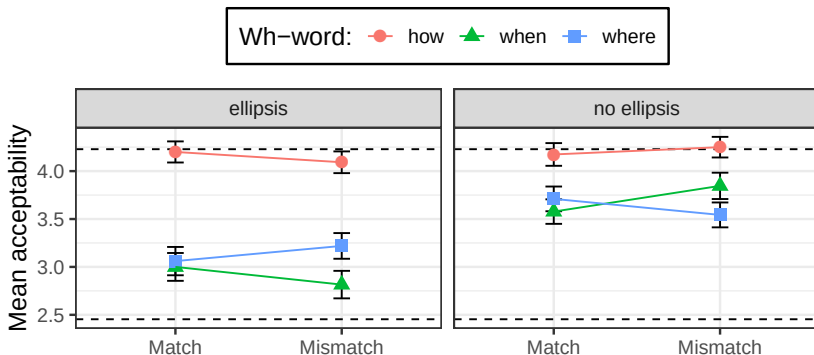
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- 24 items like (18)
- 2x2x3 design:
 - mismatch vs. no mismatch
 - ellipsis vs. no ellipsis
 - how vs. **when** vs. **where**

Results (Expt 4)



- *How* sluices at ceiling.
- *When* and *where* items degraded, more so under ellipsis.
- **Main result:** No mismatch penalty anywhere.

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Caveat: no experimental data yet!

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Motivating puzzle:

(22) Joe was murdered but we don't know who #(???)

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Voice mismatch explanation

b. #(he was murdered **by**)

"No New Words" explanation

⇒ both explanations derived from IDENTITY theory of ellipsis

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"No New Words" explanation

⇒ both explanations derived from IDENTITY theory of ellipsis

Eliding "New Words" can be OK:

(23) Can I borrow your textbook. —Why?

Conclusion

Motivating puzzle:

(22) Joe was murdered but we don't know who #(???)

a. #(murdered him)

Voice mismatch explanation

b. #(he was murdered **by**)

"No New Words" explanation

⇒ both explanations derived from IDENTITY theory of ellipsis

Eliding "New Words" can be OK:

(23) Can I borrow your textbook. —Why?

Eliding voice-mismatched questions can be OK:

(24) Republicans need to be persuaded but nobody really knows how.

Partee's marbles

On a referential theory:

(25) Joe was murdered but we don't know who ϕ .

Partee's marbles

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Other pro-forms show similar behavior (“Partee's marbles”):

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- (26) a. I dropped 10 marbles and I found all but 1 of them. It has to be under the sofa.
b. I dropped 10 marbles and I only found 9 of them. #It has to be under the sofa.

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Other pro-forms show similar behavior (“Partee's marbles”):

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But:

(27) When Joe was kidnapped, they kept him in the dark for days.

Results (Expt 1): examples of “New Words” in paraphrases

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Classic cases

(28) Joe is jealous. —Do we know who?

- a. Joe is jealous of
- b. he is jealous of
- c. Why is Joe jealous
- d. Joe is specifically jealous of

Results (Expt 1): examples of “New Words” in paraphrases

Classic cases

(28) Joe is jealous. —Do we know who?

- a. Joe is jealous of
- b. he is jealous of
- c. Why is Joe jealous
- d. Joe is specifically jealous of

“Inference” items

(29) Can I borrow your textbook over the weekend? —Why?

- a. do you want to borrow it
- b. do you need it
- c. do you want my textbook next weekend
- d. Why would A like to borrow the textbook over the weekend?

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